Assignment 5

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Title:

Regression Analysis on Real-World Datasets Using Linear, Ridge, Lasso, Logistic and Multiple Regression Models
1] To predict Uber ride fare using regression models.
2] To perform statistical and regression analysis on the Pima Indians Diabetes dataset.
3] To evaluate machine learning models using metrics like R^2, RMSE and Accuracy.

Theory:

Regression Analysis

Regression analysis is a statistical method used to examine the relationship between a dependent variable and one or more independent variables. It is widely used for prediction and forecasting.

Types of Regression Used

1. Linear Regression

Linear regression models the relationship between a dependent variable and independent variables using a linear equation.

$$y = b_0 + b_1x_1 + b_2x_2 + ... + b_nx_n$$

Where
y = dependent variable
x = independent variables
b = regression coefficients

2. Ridge Regression

Ridge regression is a regularized version of linear regression that adds an **L2 penalty** to prevent overfitting.

3. Lasso Regression

Lasso regression uses **L1 regularization** and can shrink some coefficients to zero, performing feature selection.

4. Logistic Regression

Logistic regression is used for **binary classification problems**. It predicts probabilities between 0 and 1 using the logistic function.

Evaluation Metrics
R^2 Score

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

It measures how well the model explains the variance.

RMSE (Root Mean Squared Error)

$$RMSE = \sqrt{\frac{\sum(y_{pred} - y_{actual})^2}{n}}$$

Lower RMSE indicates better model performance.

Part A: Uber Fare Prediction

Dataset

Uber Fare Dataset
https://www.kaggle.com/datasets/yasserh/uber-fares-dataset

This dataset contains:

- Pickup coordinates
- Drop-off coordinates
- Passenger count
- Ride timestamp
- Fare amount

Steps Performed

1. Data Preprocessing

- Removed unnecessary columns (`key`, `Unnamed:0`)
- Removed missing values
- Converted pickup datetime into hour, day, month
- Removed invalid passenger counts
- Created **distance feature using Haversine formula**

2. Outlier Detection

Outliers were detected using **boxplots** for:

- Fare amount
- Distance
- Passenger count

Extreme values such as fares above 100 and distances above 100 km were removed.

3. Correlation Analysis

A **heatmap** was used to analyze relationships between variables.

Important observation:

- Fare amount strongly correlates with **distance**.

4. Models Implemented

1. Linear Regression
2. Ridge Regression
3. Lasso Regression

5. Model Evaluation

Model	R^2 Score	RMSE
Linear Regression	0.766	4.44
Ridge Regression	0.766	4.44
Lasso Regression	0.765	4.45

Observation

Ridge regression performed slightly better due to regularization.

Part B: Diabetes Dataset Analysis

Dataset

Pima Indians Diabetes Dataset
https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database

Features include:

- Pregnancies
- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function
- Age
- Outcome (0 = No diabetes, 1 = Diabetes)

1. Univariate Analysis

Univariate analysis studies **one variable at a time**.

Statistical measures calculated:

- Frequency
- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Skewness
- Kurtosis

Example observation:

- Glucose has higher mean values for diabetic patients

- BMI distribution is positively skewed

2. Bivariate Analysis

Bivariate analysis studies the relationship between **two variables**.

Techniques used

- Correlation heatmap
- Linear regression
- Logistic regression

Observation:

- Glucose and BMI show strong relation with diabetes outcome.

3. Multiple Regression Analysis

Multiple regression was used to predict diabetes outcome using all independent variables.

Independent variables:

- Pregnancies
- Glucose
- Blood Pressure
- BMI
- Age
- Insulin

Dependent variable:

- Outcome

Model Results

Model	Result
Linear Regression	$R^2 \approx 0.30$
Logistic Regression	Accuracy ≈ 78%
Multiple Regression	$R^2 \approx 0.31$

Observation

Logistic regression performs better because the outcome variable is **binary (0 or 1)**.

Comparison of the Two Datasets

Aspect	Uber Dataset	Diabetes Dataset
Problem Type	Regression	Classification
Target Variable	Fare Amount	Diabetes Outcome
Best Model	Ridge Regression	Logistic Regression
Performance	$R^2 \approx 0.76$	Accuracy ≈ 78%

Challenges Faced

During the experiment, the following challenges were encountered:

- Handling missing values in the dataset.
- Removing outliers that affected model accuracy.
- Feature engineering such as calculating distance from coordinates.
- Selecting the appropriate regression model for different datasets.
- Interpreting statistical measures like skewness and kurtosis.

Conclusion

In this experiment, regression analysis techniques were applied to two real-world datasets. For the Uber dataset, Linear, Ridge, and Lasso regression models were implemented to predict ride fare. After preprocessing and feature engineering, Ridge regression produced the best results with an R^2 score of approximately 0.76.

For the diabetes dataset, statistical analysis and regression techniques were applied. Logistic regression performed better for predicting diabetes outcomes due to the binary nature of the target variable. The experiment demonstrated how different regression techniques can be used for prediction and analysis depending on the nature of the dataset.

Program:

predictUber.py ✕ predictDiabetes.py

predictUber.py > ...

```python
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.preprocessing import StandardScaler

from math import radians, cos, sin, asin, sqrt

df = pd.read_csv("uber.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Info:")
print(df.info())

df = df.drop(['Unnamed: 0','key'], axis=1)

df = df.dropna()

df = df[df['passenger_count'] > 0]
df = df[df['passenger_count'] <= 6]

df['pickup_datetime'] = pd.to_datetime(df['pickup_datetime'])

df['hour'] = df['pickup_datetime'].dt.hour
df['day'] = df['pickup_datetime'].dt.day
df['month'] = df['pickup_datetime'].dt.month
df['year'] = df['pickup_datetime'].dt.year

df = df.drop('pickup_datetime', axis=1)

```

predictUber.py > ...

```python
def haversine(lon1, lat1, lon2, lat2):

    lon1, lat1, lon2, lat2 = map(radians,
                                 [lon1, lat1, lon2, lat2])

    dlon = lon2 - lon1
    dlat = lat2 - lat1

    a = sin(dlat/2)**2 + cos(lat1)*cos(lat2)*sin(dlon/2)**2
    c = 2 * asin(sqrt(a))

    r = 6371
    return c * r

df['distance_km'] = df.apply(lambda row:
    haversine(row['pickup_longitude'],
              row['pickup_latitude'],
              row['dropoff_longitude'],
              row['dropoff_latitude']), axis=1)

df = df[(df['fare_amount'] > 0) & (df['fare_amount'] < 100)]


df = df[(df['distance_km'] > 0) & (df['distance_km'] < 100)]

plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix")
plt.show()

plt.figure(figsize=(6,4))
sns.boxplot(x=df['fare_amount'])
plt.title("Outliers in Fare Amount")
plt.show()


plt.figure(figsize=(6,4))
```

predictUber.py > ...

```python
    X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

ridge = Ridge(alpha=1.0)
ridge.fit(X_train, y_train)

y_pred_ridge = ridge.predict(X_test)

lasso = Lasso(alpha=0.1)
lasso.fit(X_train, y_train)

y_pred_lasso = lasso.predict(X_test)

def evaluate_model(name, y_test, y_pred):

    r2 = r2_score(y_test, y_pred)
    rmse = np.sqrt(mean_squared_error(y_test, y_pred))

    print(name)
    print("R2 Score:", r2)
    print("RMSE:", rmse)

evaluate_model("Linear Regression", y_test, y_pred_lr)

evaluate_model("Ridge Regression", y_test, y_pred_ridge)

evaluate_model("Lasso Regression", y_test, y_pred_lasso)
```

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predictDiabetes.py > ...

```python
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import accuracy_score, r2_score

df = pd.read_csv("diabetes.csv")

print("First 5 Rows")
print(df.head())

print("\nDataset Info")
print(df.info())

print("\nUNIVARIATE ANALYSIS\n")

for column in df.columns:

    print("Column:", column)

    print("Frequency:\n", df[column].value_counts().head())

    print("Mean:", df[column].mean())
    print("Median:", df[column].median())
    print("Mode:", df[column].mode()[0])
    print("Variance:", df[column].var())
    print("Standard Deviation:", df[column].std())
    print("Skewness:", df[column].skew())
    print("Kurtosis:", df[column].kurt())

    print("----------------------------")

df.hist(figsize=(12,10))
```

```
    X, y, test_size=0.2, random_state=42
)

lin_reg = LinearRegression()

lin_reg.fit(X_train, y_train)

y_pred_lin = lin_reg.predict(X_test)

r2 = r2_score(y_test, y_pred_lin)

print("\nLinear Regression R2 Score:", r2)

log_reg = LogisticRegression(max_iter=1000)

log_reg.fit(X_train, y_train)

y_pred_log = log_reg.predict(X_test)

accuracy = accuracy_score(y_test, y_pred_log)

print("Logistic Regression Accuracy:", accuracy)

multi_reg = LinearRegression()

multi_reg.fit(X_train, y_train)

y_pred_multi = multi_reg.predict(X_test)

r2_multi = r2_score(y_test, y_pred_multi)

print("Multiple Regression R2 Score:", r2_multi)

print("\nMODEL COMPARISON")
print("Linear Regression R2:", r2)
print("Multiple Regression R2:", r2_multi)
print("Logistic Regression Accuracy:", accuracy)
```

Output:

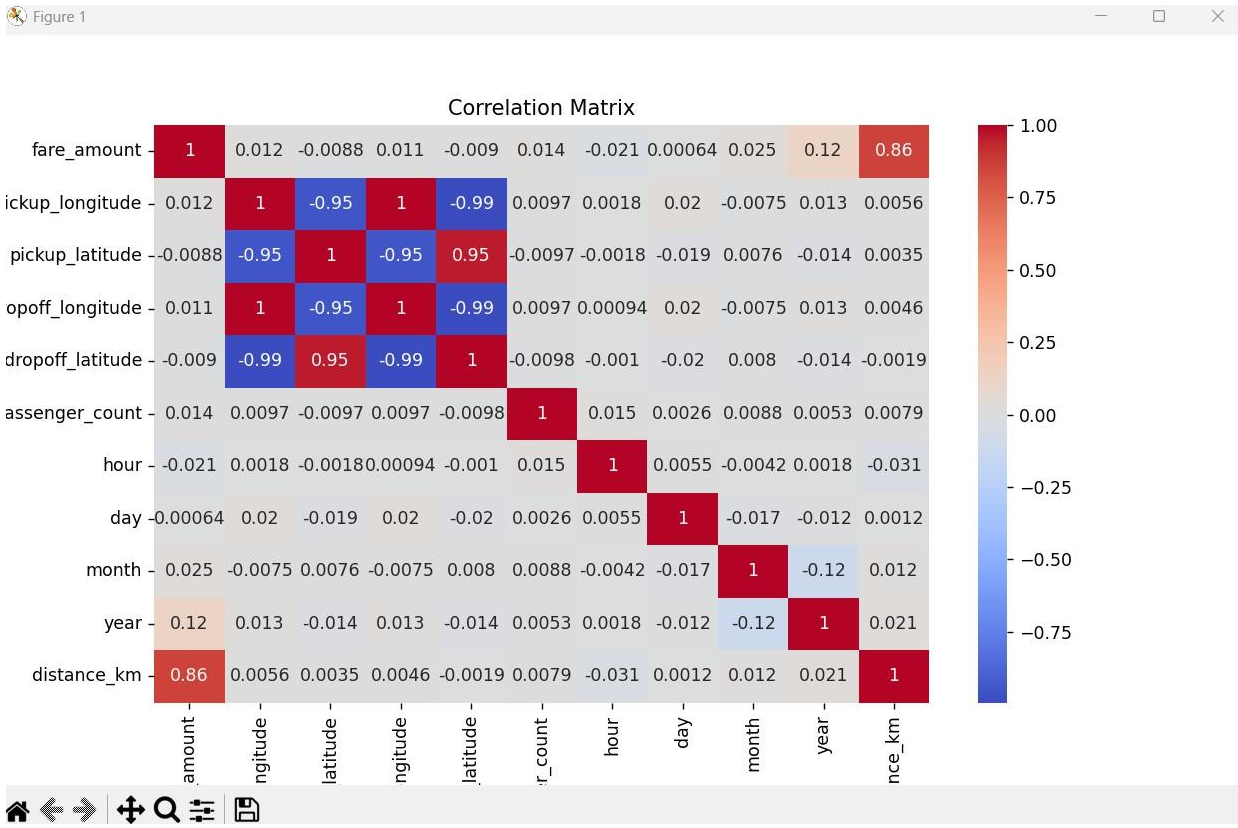


Figure 1
Correlation Matrix
fare_amount
ickup_longitude
pickup_latitude
opoff_longitude
dropoff_latitude
assenger_count
hour
day
month
year
distance_km
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

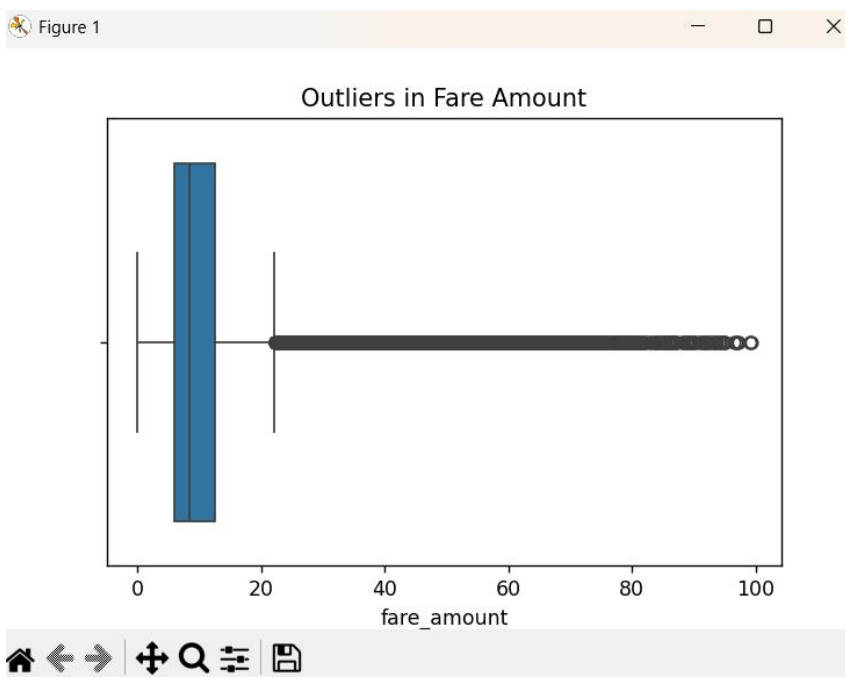


Figure 1
Outliers in Fare Amount
0
20
40
60
80
100
fare_amount

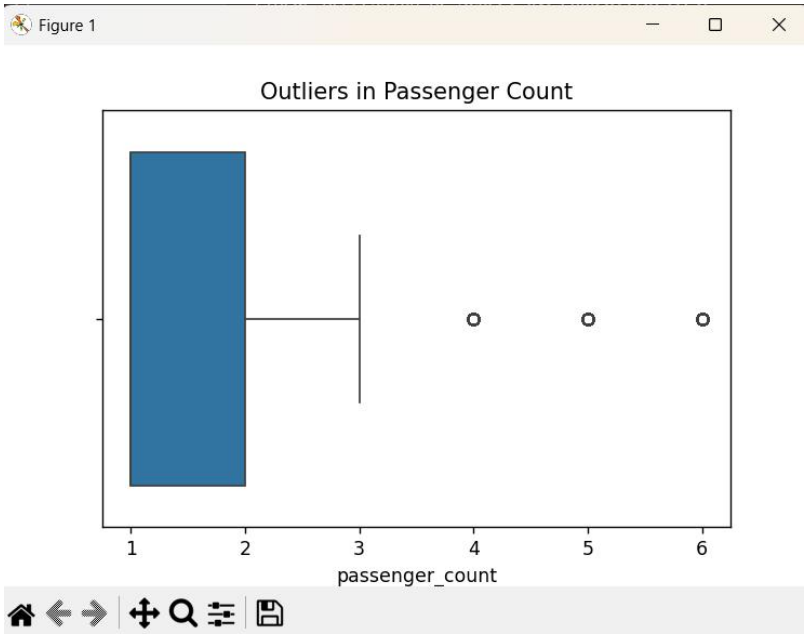


Figure 1
Outliers in Passenger Count
1
2
3
4
5
6
passenger_count

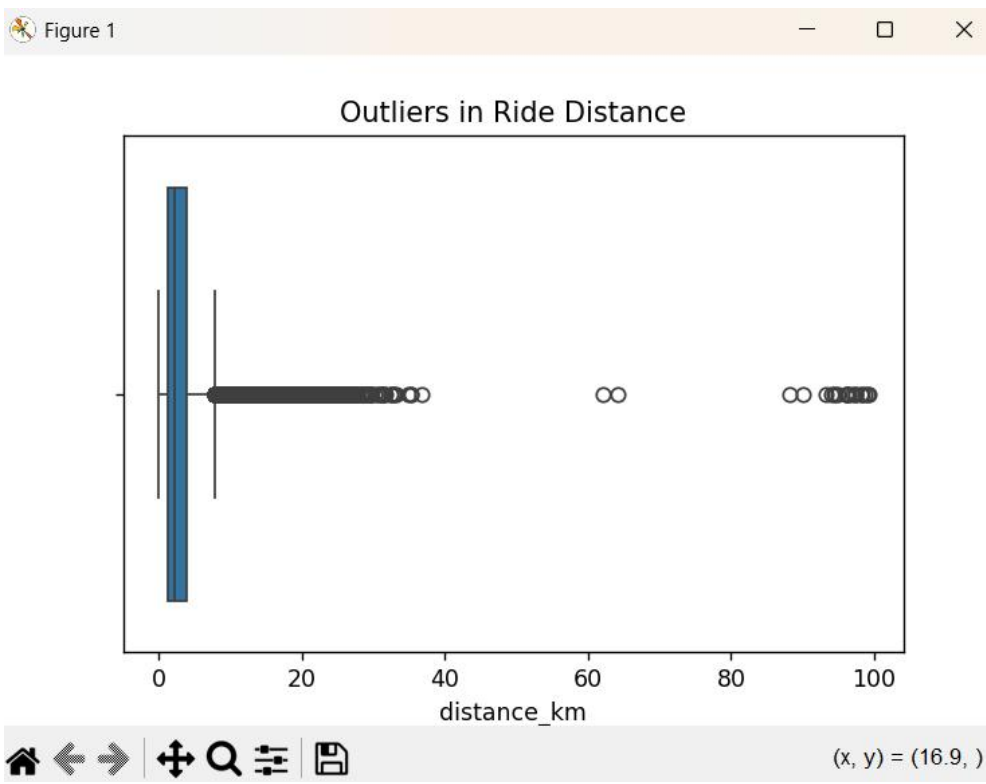
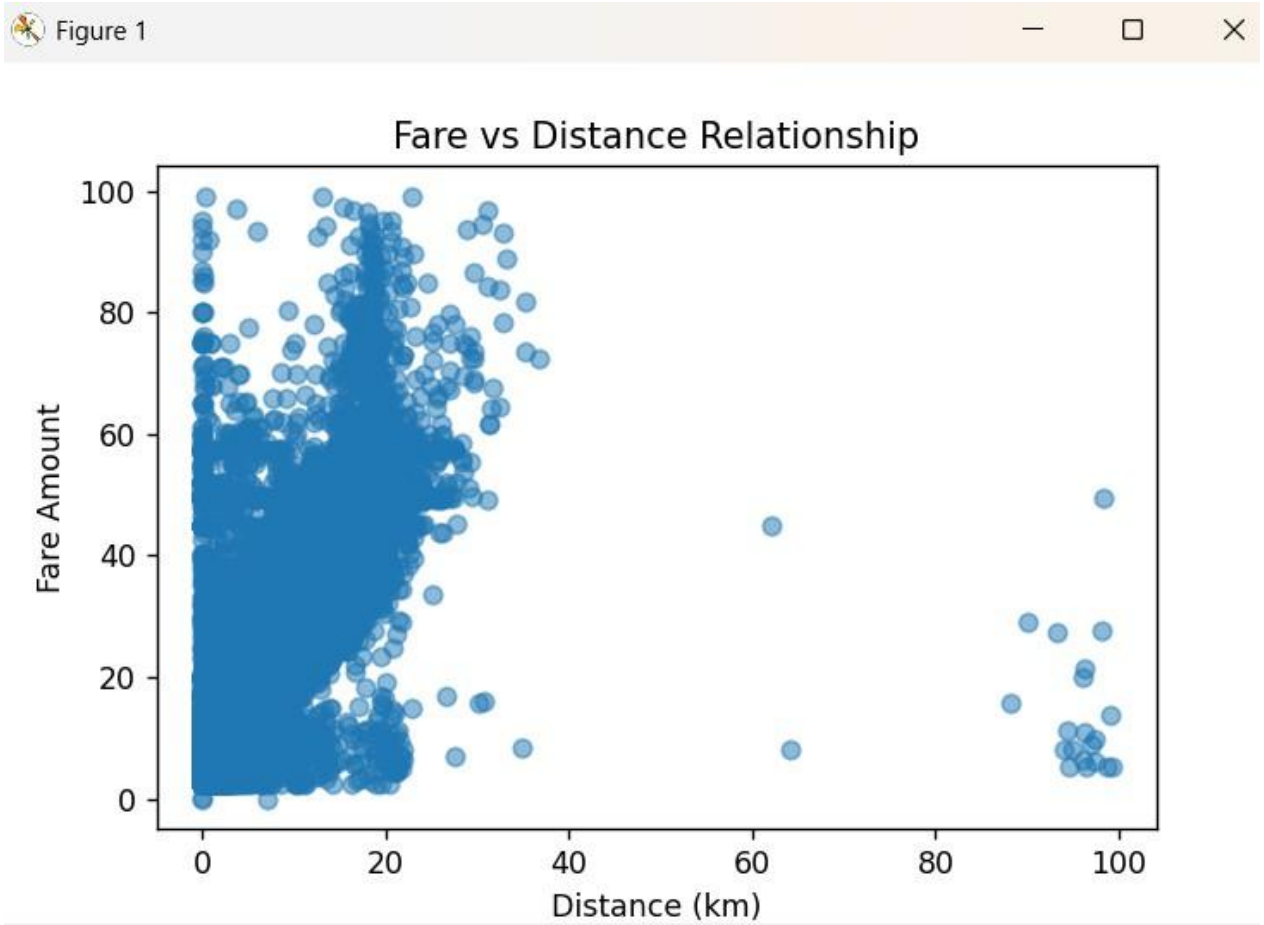


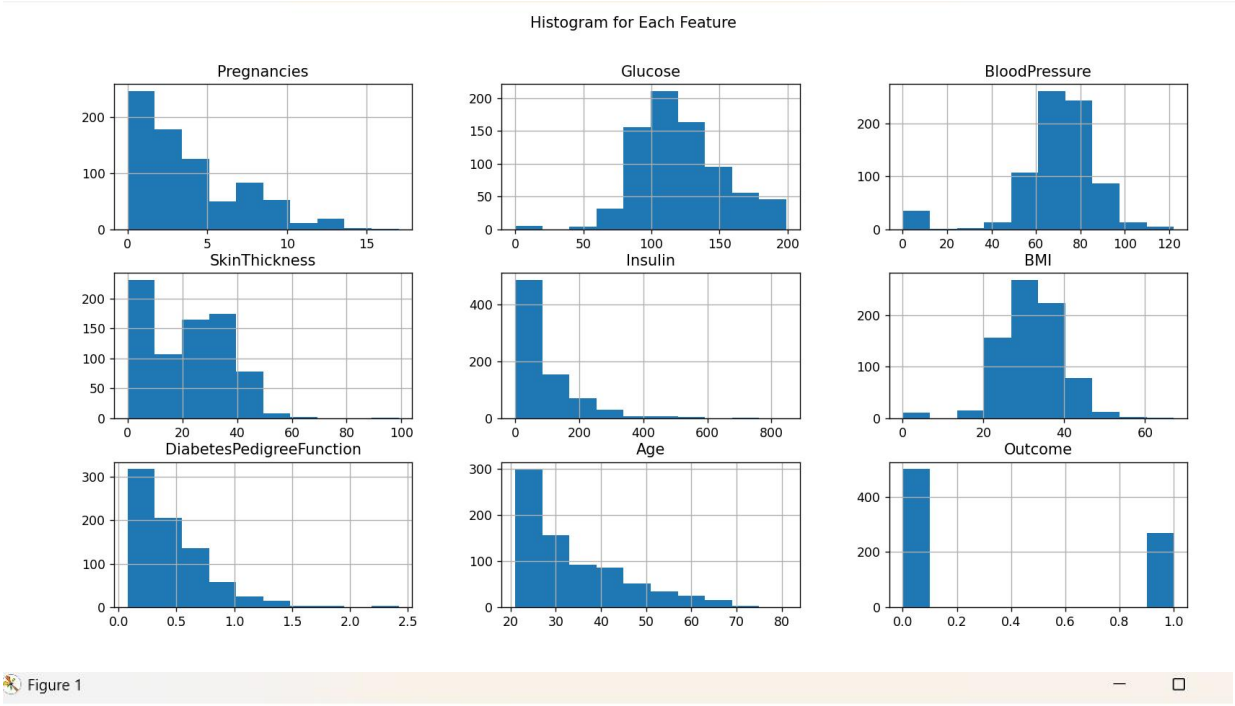
Figure 1
Outliers in Ride Distance
0
20
40
60
80
100
distance_km
(x, y) = (16.9,)

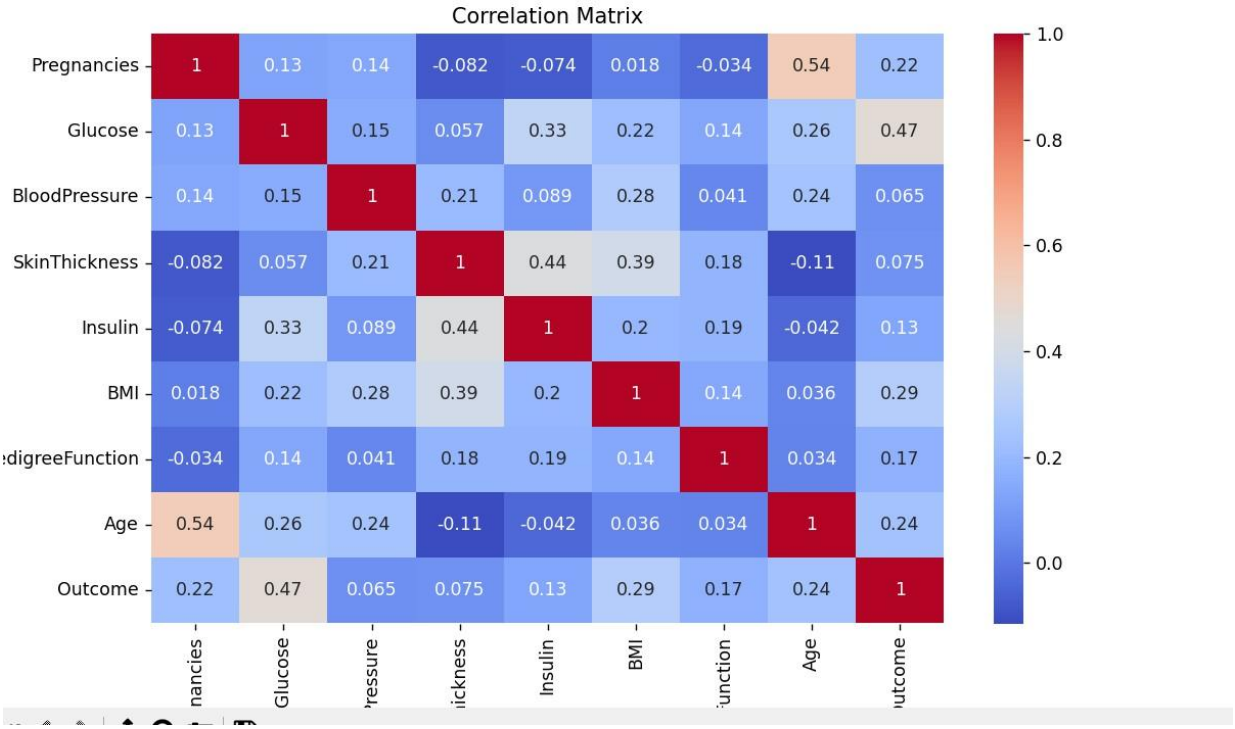


4. Models Implemented

```
Linear Regression
R2 Score: 0.7664811297836269
RMSE: 4.447902463823257
Ridge Regression
R2 Score: 0.7664807862708993
RMSE: 4.4479057353159295
Lasso Regression
R2 Score: 0.7655792362993206
RMSE: 4.456483501375775
PS D:\MLassignments\Assignment5>
```

B]


Histogram for Each Feature
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome
Figure 1


Correlation Matrix
Pregnancies 1 0.13 0.14 -0.082 -0.074 0.018 -0.034 0.54 0.22
Glucose 0.13 1 0.15 0.057 0.33 0.22 0.14 0.26 0.47
BloodPressure 0.14 0.15 1 0.21 0.089 0.28 0.041 0.24 0.065
SkinThickness -0.082 0.057 0.21 1 0.44 0.39 0.18 -0.11 0.075
Insulin -0.074 0.33 0.089 0.44 1 0.2 0.19 -0.042 0.13
BMI 0.018 0.22 0.28 0.39 0.2 1 0.14 0.036 0.29
edigreeFunction -0.034 0.14 0.041 0.18 0.19 0.14 1 0.034 0.17
Age 0.54 0.26 0.24 -0.11 -0.042 0.036 0.034 1 0.24
Outcome 0.22 0.47 0.065 0.075 0.13 0.29 0.17 0.24 1
nancies
Glucose
ressure
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Insulin
BMI
unction
Age
utcome
1.0
0.8
0.6
0.4
0.2
0.0

```
Linear Regression R2 Score: 0.25500281176741757
Logistic Regression Accuracy: 0.7467532467532467
Multiple Regression R2 Score: 0.25500281176741757

MODEL COMPARISON
Linear Regression R2: 0.25500281176741757
Multiple Regression R2: 0.25500281176741757
Logistic Regression Accuracy: 0.7467532467532467
```